

The list of experiments you've provided forms the core of a standard Analog Electronics or Analog Integrated Circuits (AIC) lab, very common across engineering universities like NSUT. Because these are standard experiments, many students and educators have uploaded their SPICE simulation files (like .asc for LTspice) to GitHub. Per your preference, I have focused on repositories that provide straight-to-the-point schematics and simulation files without excessive boilerplate code or long comments.

Here are the most relevant GitHub repositories categorized by your experiment list:

1. CMOS & IC Design (Experiments 2, 3, 5, 6)

For the MOS Cascode mirror, MOS differential amplifier, Cascode amplifier, and the two-stage CMOS op-amp, you need repositories focused on **Analog IC Design**.

- **chrisenz/Analog-Circuit-Design**: This is an excellent, highly structured repository. It contains SPICE files for common-source stages, various OTAs (Operational Transconductance Amplifiers), folded-cascode configurations, and symmetrical/telescopic op-amps.
- **Hrithik-cmd/low-noise-cmos-opamp-gpdk180**: Specifically useful for understanding the implementation of a transistor-level two-stage CMOS operational amplifier.

2. BJT Amplifiers & Current Mirrors (Experiments 1, 4, 11)

For the BJT Wilson Current Mirror, BJT differential amplifier with active load, and the Class AB push-pull amplifier, you'll want repositories that deal heavily with discrete component modeling.

- **varunnagpaal/Analog-Design-Modelling**: This repo explicitly focuses on the hands-on design, modeling, and SPICE simulation of both discrete and integrated analog circuits. It covers steady-state bias point analysis and transient analysis, which are exactly what you need to evaluate the performance of Wilson mirrors and active loads.
- **edstott/EEE1labs**: Contains dedicated LTspice laboratory sections. Part 4 specifically covers the Wilson Current mirror, explaining how to evaluate output resistance and the I_C matching differences compared to a simple mirror.

3. Oscillators, Feedback & 555 Timer (Experiments 7, 8, 9, 10, 12)

These experiments test frequency response, positive/negative feedback topologies, and transient waveform generation.

- **AriJethin/WaveForm Generator**: Contains direct LTspice .asc files for waveform generation, specifically covering Op-Amp based Wien-bridge oscillators and square wave generators.
- **Dulina14/Analog-Inductance-and-Capacitance-Measuring-Meter**: Contains LTspice simulation files for the Colpitts Oscillator (using 2N2222 transistors) and other multi-vibrator setups which act as a great baseline for your Hartley and Colpitts requirements.
- **sherlock0019/Electronics-Lab**: This is an IIT Dharwad analog lab repository. It's a great reference for standard university lab setups, including negative feedback amplifier topologies (series-series, series-parallel, etc.) and RC phase shift oscillators.

A quick tip for proceeding in LTspice: Most of the MOS-based experiments (like the Cascode or CMOS op-amp) will require you to include a specific technology parameter library (like .model or .lib files for TSMC 180nm or 90nm). Make sure to download the specific .txt or .lib parameter

files if a repository includes them alongside the .asc schematic file, otherwise, the simulation won't run.